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Example (5):

A 2200/200 Volt's Transformer
take 0.5 Ampere at a p.f. of 0.3,
on o/p circuit. Find:

- (a) the magnetising current.
- (b) the iron loss current.
- (c) K ratio.

$$V_1 = 2200 \text{ V} \quad V_2 = 200 \text{ V}$$

$$I_0 = 0.5 \text{ A} \quad \text{p.f.} = 0.3$$

$$I_m = ? \quad I_w = ?$$

$$I_w = I_0 \cos \theta$$

$$\text{p.f.} = \cos \theta = 0.3$$

$$I_w = 0.5 \times 0.3$$

$$I_w = 0.15 \text{ Ampere}$$

$$\cos \theta = 0.3$$

$$\theta = \cos^{-1}(0.3) = 72.5^\circ$$

$$I_m = I_0 \sin \theta$$

$$= 0.5 \sin 72.5^\circ$$

$$I_m = 0.4769 \text{ Ampere}$$

$$K = \frac{V_2}{V_1} = \frac{200}{2200}$$

$$K = 0.09$$

Transformer

$$N_1 = 48 \text{ turns} \quad N_2 = 12 \text{ turns}$$

$$R_L = 50 \Omega \quad V_1 = 16 \text{ V}$$

$$R_{in} = ? \quad I_1 = ? \quad \odot I_1 \rightarrow I_2$$

$$R_{in} = \frac{V_i}{I_i} = \frac{V_1}{I_1}$$

$$V_2 = \frac{I_2}{2} R_L$$

$$\frac{I_2}{2} = \frac{V_2}{R_L} =$$

$$\frac{N_2}{N_1} = \frac{V_2}{V_1}$$

$$\frac{12}{48} = \frac{V_2}{16}$$

$$V_2 = \frac{12}{48} \times 16$$

$$V_2 = 4 \text{ Volt}$$

$$\frac{N_2}{N_1} = \frac{I_1}{I_2}$$

$$I_1 = \frac{N_2}{N_1} \cdot I_2 = \frac{12}{48} \times 0.08$$

$$I_1 = 0.02 \text{ A}$$

$$I_2 = \frac{V_2}{R_L} = \frac{4}{50}$$

$$I_2 = 0.08 \text{ A}$$

$$\therefore R_{in} = \frac{V_1}{I_1} = \frac{16}{0.02}$$

$$R_{in} = 800 \Omega$$

$$\frac{V_2}{V_1} = \frac{I_1}{I_2}$$

$$I_1 = \frac{V_2}{V_1} \cdot \frac{I_2}{2} = \frac{4}{16} \times 0.08 = 0.02 \text{ A}$$

Example 4:

The transformer has 48 turns on its primary winding and 12 turns on its secondary winding. A 50Ω load is connected across the secondary. The magnitude of primary voltage is 16 V.

- What is the input resistance of the transformer.
- Use the input resistance calculate to determine the magnitude of the primary current.
- Find the magnitude of the primary current by finding i_2 .

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Transformer

$$N_1 = 48 \text{ turns} \quad N_2 = 12 \text{ turns}$$

$$R_L = 50 \Omega \quad V_1 = 16 \text{ V}$$

$$R_{in} = ? \quad I_1 = ? \quad \textcircled{I_1 \rightarrow I_2}$$

$$R_{in} = \frac{V_i}{I_i} = \frac{V_1}{I_1}$$

$$V_2 = \frac{I_2 R_L}{2}$$

$$\frac{I_2}{2} = \frac{V_2}{R_L}$$

$$\frac{N_2}{N_1} = \frac{V_2}{V_1}$$

$$\frac{12}{48} = \frac{V_2}{16}$$

$$V_2 = \frac{12}{48} \times 16$$

$$V_2 = 4 \text{ Volts}$$

$$\frac{N_2}{N_1} = \frac{I_1}{I_2}$$

$$\frac{I_1}{1} = \frac{N_2}{N_1} \cdot I_2 = \frac{12}{48} \times 0.08$$

$$\frac{I_1}{1} = 0.02 \text{ A}$$

$$I_2 = \frac{V_2}{R_L} = \frac{4}{50}$$

$$I_2 = 0.08 \text{ A}$$

$$R_{in} = \frac{V_1}{I_1} = \frac{16}{0.02}$$

$$R_{in} = 800 \Omega$$

$$\frac{V_2}{V_1} = \frac{I_1}{I_2}$$

$$I_1 = \frac{V_2}{V_1} \cdot \frac{I_2}{2} = \frac{4}{16} \times 0.08 = 0.02 \text{ A}$$

Example (4):

The transformer has 48 turns on its primary winding and 12 turns on its secondary winding. A 50Ω load is connected across the secondary. The magnitude of primary voltage is 16 V.

- what is the input resistance of the transformer.
- use the input resistance calculate to determine the magnitude of the primary current.
- Find the magnitude of the primary current by finding

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Transformer

Example (3):

A certain transformer has 50 turns on the secondary winding and 20 turns in the primary winding. An a.c. voltage having a peak value 12 V is

connected across the primary winding.
(a) what is the turn ratio of the transformer.

(b) what is the effective value of the secondary voltage?

$$N_2 = 50; \quad N_1 = 20 \text{ turns}$$

$$V_1 = 12 \text{ V}$$

(a) $k = ?$ (b) $V_2 = ?$
r.m.s.

$$(b) \frac{N_2}{N_1} = \frac{V_2}{V_1}$$

$$V_2 = \frac{N_2}{N_1} \cdot V_1$$

$$= \frac{50}{20} \cdot 12$$

$$V_2 = 30 \text{ V}$$

$$V_{2 \text{ rms}} = \frac{V_{2 \text{ max}}}{\sqrt{2}} = \frac{30}{\sqrt{2}} = 21.21 \text{ V}$$

$$(a) k = \frac{N_2}{N_1} = \frac{50}{20}$$

$$k = 2.5$$

Example (4):

The transformer has 50 turns on the secondary winding and 20 turns on the primary winding. An a.c. voltage having a peak value 12 V is connected across the primary winding.

(a) what is the turn ratio of the transformer?

(b) use the transformer ratio to find the effective value of the secondary voltage.

(c) Find the effective value of the primary voltage.

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Transformer

Example (2).

The no-load current of a transformer is 4A at 0.25 p.f. when supplied at 250V, 50 Hz. The number of turns on the primary winding is 200. Calculate:

- r.m.s. values of the flux in the core.
- the core loss
- the magnetizing component of the no-load current.

$$I_0 = 4A \quad \text{p.f.} = 0.25 \quad N_1 = 200$$

$$V_1 = 250V \quad f = 50 \text{ Hz}$$

$$\phi_{\text{r.m.s}} = ? \quad I_w = ? \quad I_m = ?$$

$$E_1 = 4.44 f N_1 \phi_{\text{max}}$$

$$\phi_{\text{max}} = \frac{E}{4.44 f N_1}$$

$$= \frac{250}{4.44 \times 50 \times 200}$$

$$\phi_{\text{max}} = 5.63 \times 10^{-3} \text{ wb}$$

$$\phi_{\text{r.m.s}} = \frac{\phi_m}{\sqrt{2}} = \frac{5.63 \times 10^{-3}}{\sqrt{2}}$$

$$\phi_{\text{r.m.s}} = 3.98 \times 10^{-3} \text{ wb}$$

$$\text{Core loss } I_w = I_0 \cos \theta$$

$$\cos \theta = \text{p.f.} = 0.25$$

$$= 4 \times 0.25$$

$$I_w = 1 \text{ Ampere}$$

$$\frac{I}{m} = \frac{I}{0} \sin \theta$$

$$\text{p.f.} = \cos \theta = 0.25$$

$$\theta = \cos^{-1}(0.25) = 75.52^\circ$$

$$I_m = 4 \times \sin(75.52^\circ)$$

$$I_m = 3.87 \text{ Ampere}$$

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Transformer

Example (1):

A single-phase transformer is required to supply 15 A at 63 V when operating from 200 V, 50 Hz mains.

The core area is 9.677 cm^2 and the maximum flux density is 0.867 Wb/m^2 .

Neglecting losses and effective of magnetizing component determine:

(a) the supply current.

(b) the number of turns on primary and secondary windings.

$$I_2 = 15 \text{ A} \quad V_2 = 63 \text{ V} \quad A = 9.677 \text{ cm}^2 = 9.677 \times 10^{-4} \text{ m}^2$$

$$V_1 = 200 \text{ V} \quad f = 50 \text{ Hz}$$

$$B = 0.867 \text{ Wb/m}^2$$

$$I_1 = ? \quad N_1 = ? \quad N_2 = ?$$

$$k = \frac{N_2}{N_1} = \frac{V_2}{V_1} = \frac{I_1}{I_2}$$

$$\frac{V_2}{V_1} = \frac{I_1}{I_2}$$

$$\frac{63}{200} = \frac{I_1}{15}$$

$$I_1 = \left(\frac{63}{200} \right) \times 15$$

$$I_1 = 4.725 \text{ Ampere}$$

$$E_1 = 4.44 f N_1 \Phi_m$$

$$\Phi = B \cdot A$$

$$\Phi_{\max} = 0.867 \times 9.677 \times 10^{-4}$$

$$\Phi_{\max} = 8.389 \times 10^{-4} \text{ Wb}$$

$$N_1 = \frac{E_1}{4.44 f \Phi_m}$$

$$= \frac{200}{4.44 \times 50 \times 8.389 \times 10^{-4}}$$

$$N_1 = 1074 \text{ turn}$$

$$\frac{N_2}{N_1} = \frac{V_2}{V_1}$$

$$N_2 = \frac{V_2}{V_1} \cdot N_1$$

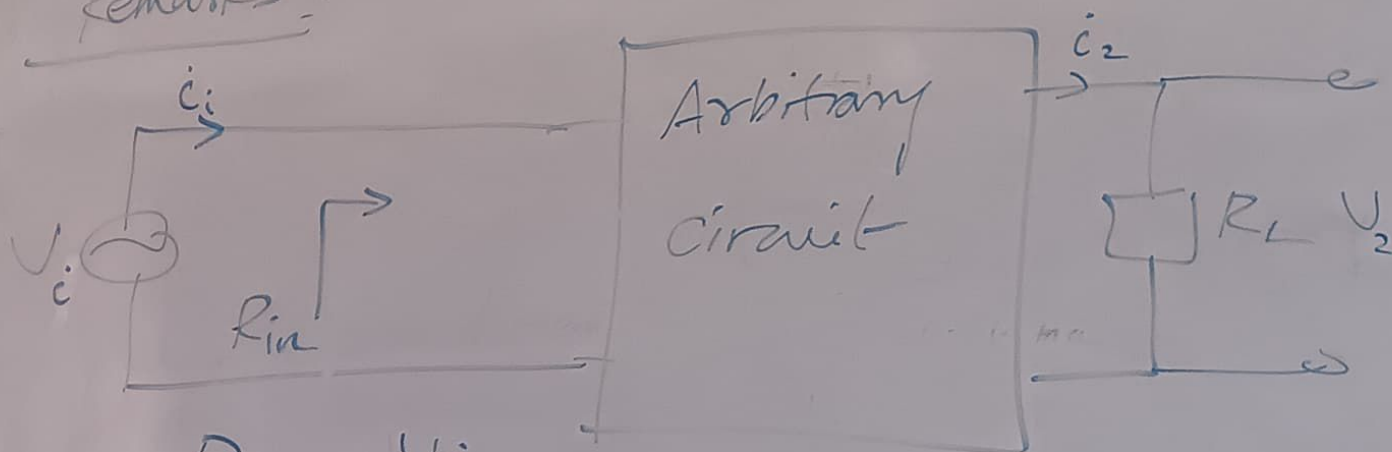
$$= \frac{63}{200} \times 1074$$

$$N_2 = 338 \text{ turns}$$

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Transformer

Remarks :-



$$R_{in} = \frac{V_i}{i_i}$$

$$R_{in} = \left(\frac{N_1}{N_2} \right) R_L$$

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Transformer

The voltage transformer ratio is given by

$$K = \frac{N_2}{N_1} = \frac{V_2}{V_1} = \frac{I_1}{I_2}$$

When the transformer is on no-load the primary draws a current of I_0 which has two components

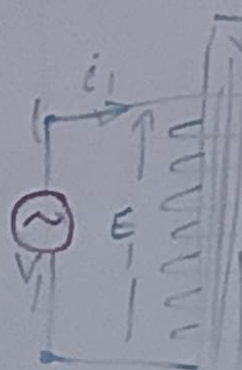
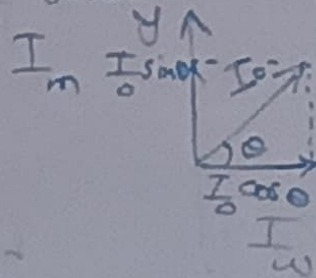
$$I_0 = \sqrt{I_w^2 + I_m^2}$$

I_m = magnetising component

$$I_m = I_0 \sin \theta$$

I_w = core loss component (Wattless component)

$$I_w = I_0 \cos \theta$$



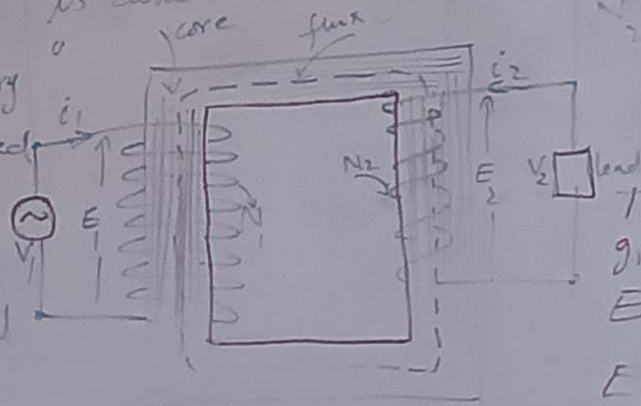
The volt
Primary
while the
is call
0

Transformer

decrease or increase in current.

A transformer consists of two winding, the primary and secondary, wound on a common laminated magnetic core. The winding connected to the a.c. source is called primary winding and the other connected to the load is called secondary winding.

The voltage in the input is called primary voltage and is denoted by V_1 , while the voltage in the output is called secondary voltage V_2 .



E_1 Induced in the primary winding
 E_2 " " " Secondary winding
 N_1 Number of turns in primary winding
 N_2 No. of turns in secondary winding
 ϕ maximum flux in the core

The emf of a transformer is given by

$$E_1 = 4.44 f N_1 \Phi_m$$

$$E_2 = 4.44 f N_2 \phi_m$$